

Development of a Machine Learning Model for Planting Time Prediction for Smallholder Farmers in The Gambia Using Rainfall Data



Project Proposal By:

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A Project Proposal Submitted for CPS 371: Artificial Intelligence

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Option A: Enhancing an Existing AI Model

Project Option: This project follows Option A. We take the Random Forest framework developed by Jeong et al. (2016) and adapt it in several important ways: we change the task from yield prediction to planting suitability classification, focus it specifically on The Gambia, engineer new climate features suited to the local context, and build a working Streamlit app that farmers or agricultural officers can actually use. None of these extensions were part of the original work.

1. Introduction

Farming is the backbone of life in The Gambia. Around 75% of the population depends on it for their income and food, and almost all of that farming relies entirely on rain rather than irrigation. When the rains arrive at the wrong time or fall short of what is needed, the consequences for families are felt straight away. The sector is particularly sensitive to climate variability — particularly rainfall timing and distribution.

Over the past few decades, rainfall in West Africa has become harder to predict. The rains now start later some years, stop earlier in others, and dry spells appear in the middle of what should be the wet season. For farmers trying to decide when to plant, this uncertainty is a real problem. Plant too soon and the soil is too dry for seeds to survive. Plant too late and the growing season is cut short before the crop can fully mature.

Farmers have always relied on traditional signs and local knowledge to decide when to plant, and for generations that worked well. But those patterns are shifting, and the old rules are becoming less dependable. This project explores whether machine learning, specifically the Random Forest method, can help fill that gap. We take an approach that has already been shown to work for crop prediction research and adapt it to answer a more practical question for Gambian farmers: is this a good time to plant?

2. Problem Statement

Farmers in The Gambia do not currently have access to any tool that can tell them, based on actual climate data, whether conditions are good for planting. Most decisions still rely on personal experience and traditional signs, which worked well in the past but are becoming less trustworthy as the climate changes. Getting the timing wrong has real consequences. Plant too early and the seeds may die because there is not enough moisture. Plant too late and the crops do not have enough time to grow before the season ends. Either way, families end up with less food and less income.

Research has shown that machine learning can do a good job of modelling how crops respond to climate. But most of that research predicts how much a crop will yield, not whether now is a good time to plant. It is also usually done at a global or regional scale, which means it does not reflect the specific conditions in a place like The Gambia. And almost none of it is packaged into a tool that a farmer or agricultural officer could

actually open and use. This project tries to address all three of those shortcomings, as described further below for real-world users. This project addresses all three gaps by developing a localised, deployable AI system that predicts planting suitability and optimal planting windows for farmers in The Gambia.

3. Related Work and Baseline Model

A growing body of literature applies machine learning to agricultural prediction under climate variability. Three bodies of work are particularly relevant to this project.

3.1 Primary Baseline — Jeong et al. (2016)

The starting point for our methodology is Jeong et al. (2016), who published “Random Forests for Global and Regional Crop Yield Predictions” in PLOS ONE. They used Random Forest models to predict crop yields for wheat, maize, and potato across different parts of the world, with results consistently better than standard linear regression. Their prediction errors ranged from just 6 to 14 percent of the average observed yield, compared to 14 to 49 percent for linear models. This established Random Forests as a reliable technique for climate and crop modelling. One thing to be clear about is that Jeong et al. were doing regression, predicting a number (how many tons per hectare). Our project does classification, predicting a category (suitable or not suitable for planting). Because of this difference, the performance scores from their study cannot be directly compared to ours. We use their framework as a methodological foundation and a way to verify our pipeline is set up correctly, not as a benchmark to beat. The authors applied Random Forest models to predict continuous crop yield values (tons/ha) in response to climate and biophysical variables at global and regional scales for wheat, maize, and potato, benchmarking against multiple linear regression (MLR). RF significantly outperformed MLR in all test cases, achieving root mean square errors of 6–14% of average observed yield compared to 14–49% for MLR. This establishes RF as a well-validated, reproducible framework for crop-climate modelling. It is important to note that Jeong et al.’s metrics (RMSE, Nash-Sutcliffe EF, Willmott’s d) apply to a regression task and are therefore not directly comparable to the classification metrics (accuracy, F1-score, AUC) used in this project. The Jeong et al. pipeline serves as methodological inspiration and a pipeline validity check, not as a performance benchmark against which our classifier results are measured.

Beyond the task difference, the original model was built using global data and was never designed to help farmers make day-to-day decisions. It has no user interface and no connection to local conditions in West Africa. Those are precisely the gaps we are trying to fill.

3.2 West Africa Monsoon Prediction — Boboceca and Atchadé (2025)

Boboceca and Atchadé (2025) used machine learning to predict when the West African monsoon season begins and when dry spells are likely to occur. Their finding that monsoon onset can be predicted with reasonable accuracy using statistical models is encouraging and supports the general approach we are taking in this project. The key differences are that they used sea surface temperatures as their inputs and a simpler logistic regression model, whereas we use local daily weather records from NASA POWER and Random Forest. They also did not build a practical tool from their findings.

3.3 West African Crop Yield Modelling — Cedric et al. (2022)

Cedric et al. (2022) predicted crop yields across several West African countries using climate and soil data, which makes their work geographically closer to ours than Jeong et al. But like Jeong et al., they focused on how much a crop would yield rather than when to plant, and their analysis worked at the country level rather than looking at specific farming regions within a country. Our project differs in that it targets individual agricultural zones within The Gambia, asks a planting timing question rather than a yield question, and produces a usable application. Their work operates at country level rather than the sub-national scale needed for localised planting guidance, and it retains the yield regression framing rather than producing planting recommendations. Our work is distinguished by its classification approach, Gambia-specific localisation, and accessible deployment.

Enhancement Summary: Starting point: Jeong et al. (2016) Random Forest for crop yield regression, with no deployment component. Direct performance benchmark for our classifier: a logistic regression model trained on the same Gambia dataset. What we add: (1) a new task — predicting planting suitability rather than yield; (2) localisation to The Gambia using NASA POWER data; (3) new features designed around rainfall onset patterns specific to West African agriculture; (4) a working Streamlit application that delivers results to users.

4. Objectives

4.1 Main Objective

To take an established Random Forest crop-climate model and develop it into a practical planting suitability prediction tool for The Gambia, trained on 30 years of NASA POWER climate data and delivered through a simple interactive application that anyone can use without a technical background.

4.2 Specific Objectives

- Reproduce the Random Forest pipeline from Jeong et al. (2016) as a yield regression task and record its error metrics to confirm that our implementation is working correctly. Then train a logistic regression classifier on our Gambia dataset to serve as the comparison point for our main model.
- Collect and process 30+ years of daily climate data from the NASA POWER API for key agricultural regions in The Gambia.
- Engineer agronomically meaningful features including rainfall onset indicators, rolling rainfall averages, seasonal temperature windows, and day-of-year encodings.
- Generate binary planting suitability labels using published agronomic thresholds, enabling supervised classification.
- Train the Random Forest classifier, tune its settings, and show that it performs better than the logistic regression baseline on F1-score and AUC when tested on the same Gambia dataset.
- Evaluate model performance using accuracy, precision, recall, F1-score, and AUC.
- Develop an interactive Streamlit prototype for user-based planting predictions.

5. Methodology

5.1 Pipeline Verification — Jeong et al. (2016) Reproduction

We start by rebuilding the Random Forest pipeline from Jeong et al. (2016) in Python, keeping the setup as close to the original paper as we reasonably can. The model will predict yield as a number rather than a category, and we will record the standard error metrics used in the paper. This is just to check that our implementation is correct before we move on to the Gambia work. It is not meant to be a direct comparison with our classifier. For that comparison, we train a separate logistic regression model on our Gambia dataset and measure both models on the same data. A RF regressor will be trained on climate-derived features (temperature, precipitation, fertiliser, evapotranspiration) and its performance recorded using regression metrics: RMSE, Nash-Sutcliffe EF, and Willmott's d . This step serves as a pipeline validity check — confirming the RF framework is correctly implemented — rather than as a direct performance benchmark for our classifier. A separate logistic regression classifier, trained on the same Gambia-specific labelled dataset, will serve as the actual quantitative baseline against which the enhanced RF classifier is compared using classification metrics: accuracy, precision, recall, F1-score, and ROC-AUC.

5.2 Data Collection

We will download daily climate records going back to 1990 from the NASA POWER API, covering the main farming regions of The Gambia. The data includes daily rainfall, minimum and maximum temperature, relative humidity, and solar radiation. Optional soil

texture and moisture data may be sourced from the FAO Global Soil Database or ISRIC World Soil Information to supplement the climate inputs.

5.3 Data Processing and Feature Engineering

Once the raw data is downloaded, we will clean it by filling in missing values and handling any unusual readings. We will then create a set of new variables that are more meaningful for planting decisions than the raw daily numbers on their own:

- Rolling rainfall sums: 3-day, 7-day, and 30-day cumulative precipitation
- Rainfall onset indicators: based on the standard 20mm/3-consecutive-day threshold used in West African agronomic literature (Sultan & Gaetani, 2016)
- Seasonal temperature averages and day-of-year encoding for seasonality capture
- Inter-annual variability metrics to account for year-to-year climate fluctuations

5.4 Label Generation

There are no existing records in The Gambia that say whether any given period was good or bad for planting. To get around this, we create our own labels using well-established farming rules from the research literature. A period will be labelled as **Suitable (1)** if the total rainfall over consecutive days reaches the required amount and the average temperature is within the best range for the crop. If these conditions are not met, it will be marked as **Not Suitable (0)**.

5.5 Enhanced Model Training and Evaluation

With the labelled dataset ready, we will train our Random Forest classifier and tune its settings using cross-validated grid search. We will then test it on data the model has never seen and report accuracy, precision, recall, F1-score, and ROC-AUC. These results will be compared directly against the logistic regression baseline in a table, supported by confusion matrices and a chart showing which features the model found most important. As noted above, because our labels come from agronomic rules rather than actual recorded planting decisions, the evaluation measures how well Random Forest learned those rules. This is standard practice when no ground-truth records exist and will be clearly noted as a limitation in the final report.

5.6 System Design and Deployment

Once the model is trained and evaluated, we will wrap it in a Streamlit web application. A user selects their region of The Gambia from a dropdown, picks a date, and the app returns a clear answer: conditions are suitable for planting, or they are not, along with a confidence score. The application is built in a way that could be extended later to cover other countries in the region or connected to real-time weather feeds.

6. Expected Outcomes

Upon successful completion, the project will deliver the following:

- A working reproduction of the Jeong et al. (2016) pipeline with documented regression metrics confirming it runs correctly, plus a logistic regression classifier on our Gambia data that serves as the comparison model for our main results.
- A trained and tested Random Forest classifier that performs better than the logistic regression baseline on F1-score and AUC for planting suitability prediction in The Gambia.
- A working Streamlit application where a user selects their region and gets back a planting suitability result, a recommended window, and a confidence score, all driven by the trained model.
- Visualisations of seasonal rainfall trends, planting suitability windows, and model feature importances to support both user decision-making and academic reporting.
- A clean and well-documented GitHub repository that another researcher or developer could pick up and extend to other Sahelian countries or additional crops without starting from scratch.

7. Significance of the Project

What makes this project meaningful is not just that it uses machine learning, but that it redirects an existing technique towards a problem that actually matters to people on the ground. We are not predicting how much a crop will yield in a laboratory sense. We are trying to help a farmer in The Gambia decide whether to plant this week or wait. That shift in purpose, combined with localising the model to Gambian conditions and packaging the results into a usable app, is what distinguishes this work from the studies we build on. The performance comparison we make is honest and fair: our Random Forest is measured against a logistic regression model trained on exactly the same data, so the improvement we claim is real and verifiable.

On a practical level, the app gives smallholder farmers access to something they have never had before: a tool that draws on decades of climate data to support their most important seasonal decision. The interface is intentionally simple. A user does not need to understand machine learning to benefit from it. They just pick their region and get an answer. And while the project is focused on The Gambia, the same approach could be applied across the Sahel with relatively little additional work.

From an academic standpoint, the project covers the full cycle of applied machine learning: collecting and cleaning real data, engineering features, training and comparing models, evaluating results honestly, and delivering a working prototype. It is grounded in a real problem with real stakes, which is exactly what a final year capstone project should be.

8. Tools and Technologies

- Python – Primary programming language for all pipeline components.
- Pandas and NumPy – Data manipulation, cleaning, and feature engineering.
- Scikit-learn – Random Forest training, hyperparameter tuning (GridSearchCV), cross-validation, and evaluation metrics.
- Matplotlib and Seaborn – Statistical visualisations, confusion matrices, and feature importance plots.
- Streamlit – Development and deployment of the interactive user-facing prototype.
- NASA POWER API – Primary source of 30+ years of daily climate data.
- FAO / ISRIC World Soil Database – Optional supplementary soil data.
- GitHub – Version control, code documentation, and team collaboration.

9. Project Plan (4 Week Timeline)

The project is structured across four weeks for a two-person team, with one member leading the model pipeline and one leading the data processing and application. We are taking an iterative approach, making sure we have a working baseline before we move on to improvements and deployment.

1. **Week 1:** Review the literature and reproduce the Jeong et al. (2016) pipeline. Download and inspect NASA POWER climate data for each agricultural zone. Record the regression metrics from the reproduction to confirm the pipeline is working, and train the logistic regression model on the Gambia data as our comparison baseline. Record baseline performance metrics as the reference point for all subsequent comparisons.
2. **Week 2:** Clean the downloaded climate data, fill any gaps, and engineer the new features. Generate planting suitability labels using the agronomic thresholds from the literature. Run exploratory analysis and produce charts showing seasonal patterns across the different zones.
3. **Week 3:** Train the Random Forest classifier on the labelled Gambia dataset and tune its hyperparameters. Evaluate it against the logistic regression baseline and record all metrics. Produce the comparison table, confusion matrices, and feature importance charts that will go into the report.
4. **Week 4:** Build and test the Streamlit application with the trained model behind it. Write up the full report following the course structure. Prepare and rehearse the demonstration presentation.

References

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